

Neeraj Sharma

Principal AI Security Engineer | DSPM Architect | GenAI Platforms | MLSecOps

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SUMMARY

Principal-level engineer bridging 25+ years of large-scale infrastructure reliability with modern AI/ML security governance and GenAI platform engineering. Specialized in Zero Trust AI infrastructure, DSPM, and DevSecOps at scale — from IoT and global CDN platforms to cloud-native Kubernetes environments across IBM's 22+ global production data centers.

CORE SKILLS

AI/ML: GenAI, LLMs, Agentic AI, RAG, LangChain, LangGraph, RAGAS

Security: DSPM, AI Governance, Zero Trust, DevSecOps, MLSecOps, FedRAMP

Engineering: Python, REST APIs, PostgreSQL, SQL, Vector Databases

Cloud/Platform: Kubernetes, Terraform, Tekton CI/CD, IBM Cloud, GCP; AWS Bedrock (working knowledge)

WORK EXPERIENCE

Independent Consultant — AI/ML Security, DSPM & GenAI Platforms

Remote | Jan 2026 – Present

- Designed secure RAG and agentic AI solutions using Python, LangChain, LangGraph, hybrid retrieval, reranking, RAGAS, PostgreSQL/pgvector, and DSPM controls.
- Built cloud-native security and DevSecOps workflows across Kubernetes, CI/CD, REST APIs, vector databases, sensitive-data discovery, and audit evidence.

Applied AI/ML Engineering — Professional Development

Remote | Apr 2025 – Jun 2026

- Completed postgraduate programs and production-oriented projects in GenAI, DSPM, and cloud security alongside independent consulting.

IBM — San Jose, CA — Senior Backend Engineer (DevSecOps/SRE) | Apr 2021 – Apr 2025

- Evolved the backend reliability and observability stack for IBM's global network underlay infrastructure spanning 22+ production data centers worldwide, owning service-level stability across 60+ microservices.
- Hardened release pipelines using Tekton CI/CD with integrated vulnerability management and patching; authored 200+ Public Cloud Exceptions (PCEs) and led cross-functional efforts with Dev, Security, and SRE teams to reduce the PCE backlog from 200+ to 30 — an 85% reduction in cloud security debt.
- Engineered and maintained 200+ Python and Ansible automation tests, eliminating ~40 hrs/month of repeated manual regression effort and improving deployment confidence at scale.
- Operated Kafka and Zookeeper within a K8s-based observability platform to troubleshoot production incidents across 60+ microservices; leveraged ELK stack and Prometheus/Grafana to isolate root causes and maintain 99.9% alert delivery reliability.
- Earned IBM credentials in watsonx, Cloud DevSecOps, Kubernetes, Trustworthy AI, and Security & Privacy by Design.

Exabeam — San Mateo, CA — Senior Software Quality Engineer | Aug 2019 – Mar 2021

- Led validation engineering for a SIEM/UEBA platform processing billions of behavioral events daily across GCP, ensuring detection accuracy at scale for Advanced Analytics and correlated threat workflows.
- Augmented existing test frameworks with targeted event correlation tests — a key feature for filtering false positives — improving detection accuracy by 20% across distributed security analytics pipelines.
- Expanded data-driven security model coverage, reducing false-negative risk in production threat detection.

Oracle (OCI) — San Jose, CA — Principal Software Engineer | May 2018 – Jul 2019

- Architected and validated multi-tenant leaf-spine network infrastructure across 20+ global OCI data centers; delivered automation microservices that accelerated new cloud region provisioning from weeks to days.

Light.co — Palo Alto, CA — Senior Software Quality Engineer | Jun 2017 – Apr 2018

- Built end-to-end test automation for the L16 computational camera system (16-lens array), validating firmware-to-Android integration across image capture, merge, and computational photography processing pipelines.

Silver Spring Networks — Redwood City, CA — Senior Systems Integration Engineer | Nov 2012 – May 2017

- Built E2E test automation from scratch for an IoT smart grid platform supporting 23M+ deployed devices across utility networks; achieved ~60% automation coverage within a 2-week firmware release cadence, significantly reducing manual validation risk at fleet scale.

Redwood Systems — Fremont, CA — Senior Quality Engineer (SDET/Automation) | Jan 2011 – Oct 2012

- Led system integration testing for sensor-driven smart lighting platforms deployed in enterprise data centers including Meta and Microsoft facilities, validating energy management and sensor control reliability.

CDNetworks — San Jose, CA — QA Manager | Aug 2009 – Dec 2010

- Led QA strategy for a globally distributed CDN; established canary deployment validation pipelines that reduced production rollback risk and strengthened performance and data integrity across large-scale edge networks.

EDUCATION

IBM Generative AI Engineering | 2026 — DSPM-governed CVE RAG capstone using hybrid retrieval, reranking, RAGAS, and vector search.

PGP Artificial Intelligence & Deep Learning (AIDL) — UT Austin McCombs | 2026

PGP Data Science & Business Analytics (PGPDSBA) — UT Austin McCombs | 2025

Savitribai Phule Pune University — Master of Computer Management (MCM) | 1995

SELECTED AI PLATFORM & DSPM PROJECTS

- DSPM – GenAI VectorDB Governance: data asset classification, column-level sensitivity detection, audit scoring, policy enforcement, and count-based deletion verification across Chroma, Pinecone, Qdrant, Weaviate, Milvus, and OpenSearch.
- DSPM – DevSecOps Ephemeral Compute Evidence Pipeline: DSPM controls for CI/CD to detect, audit, and manage sensitive data exposure in logs, artifacts, and ephemeral environments.